Mobile Pentesting with Kali Nethunter

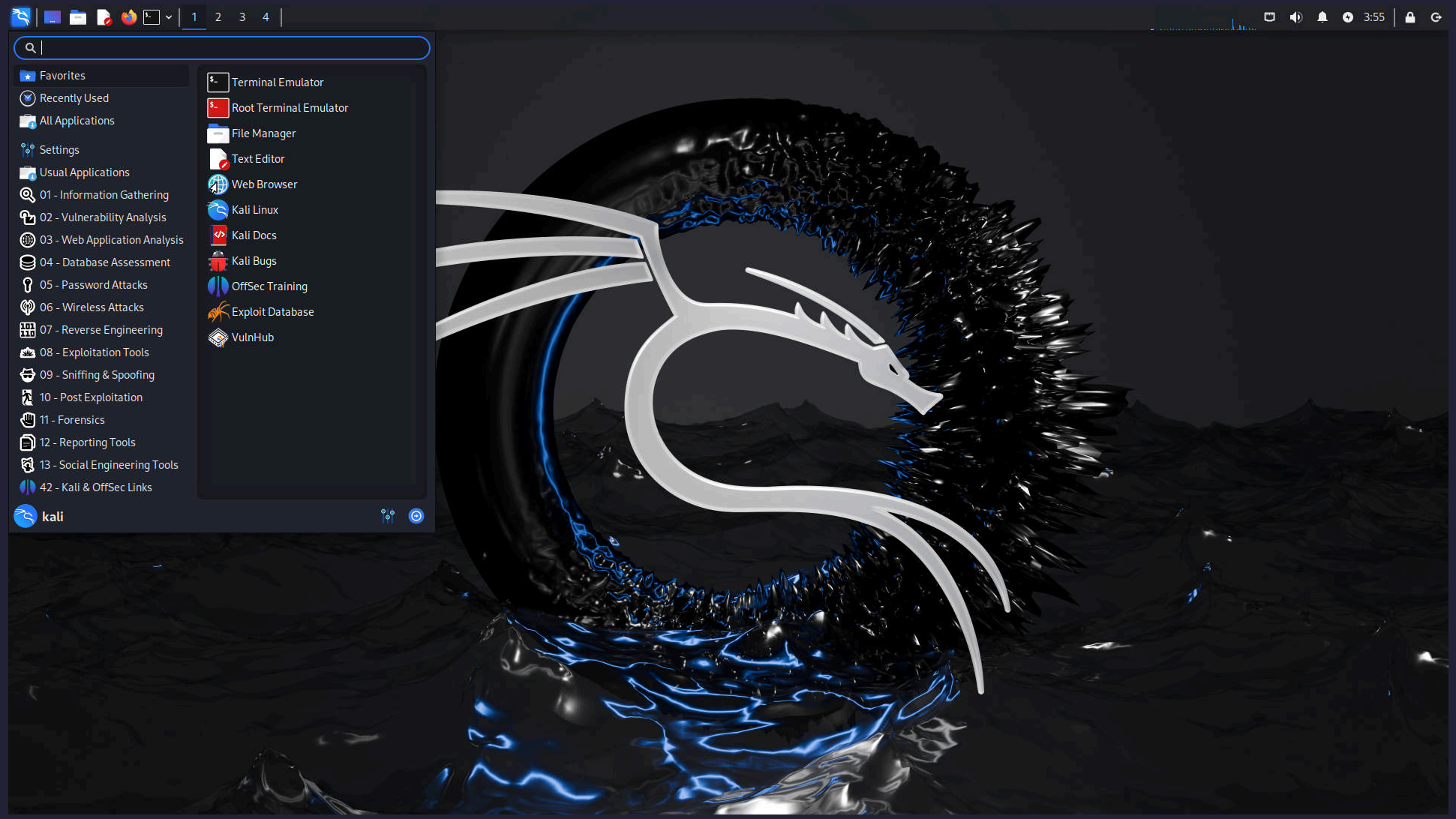
Saintcon 2025

Brayden Houston

```
$ whoami
Name: Brayden Houston / 8BitFish
Job: Cybersecurity Engineer
Interests: homelab, automation, running
Groups: DC608, Wisconsin Cyber Response Team
Extra Notes: Thoughts are my own not my employeers
$
```

Don't be evil

What is Kali?


1
2
3
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3:55
Favorites
Recently Used
All Applications
Settings
Usual Applications
01 - Information Gathering
02 - Vulnerability Analysis
03 - Web Application Analysis
04 - Database Assessment
05 - Password Attacks
06 - Wireless Attacks
07 - Reverse Engineering
08 - Exploitation Tools
09 - Sniffing & Spoofing
10 - Post Exploitation
11 - Forensics
12 - Reporting Tools
13 - Social Engineering Tools
42 - Kali & OffSec Links
Terminal Emulator
Root Terminal Emulator
File Manager
Text Editor
Web Browser
Kali Linux
Kali Docs
Kali Bugs
OffSec Training
Exploit Database
VulnHub
kali

What is Kali Nethunter?

Its Kali but on your phone

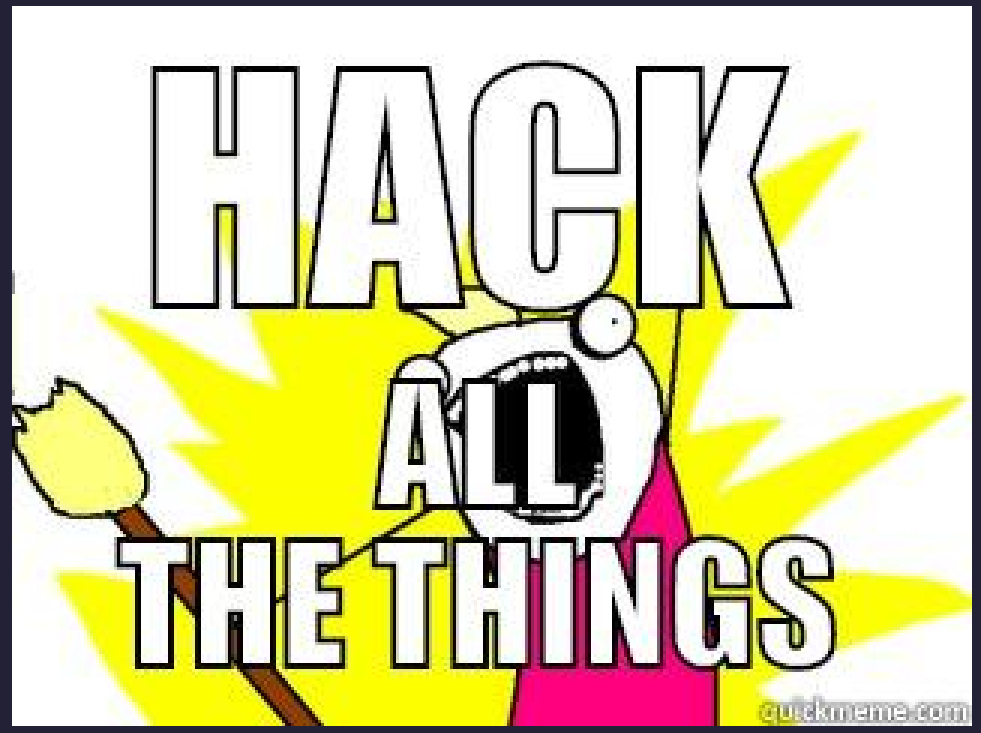
8:00
KALI
NETHUNTER

18:59
80%
NetHunter
[=] NetHunter for Termux installed successfully
[+] To start NetHunter, type:
[+] nethunter # To start NetHunter cli
[+] nethunter kex passwd # To set the KeX password
[+] nethunter kex & # To start NetHunter gui
[+] nethunter kex stop # To stop NetHunter gui
[+] nethunter -r # To run NetHunter as root
[+] nh # Shortcut for nethunter
$ nh kex &
[1] 4332
$
NetHunter KeX server sessions:
X DISPLAY # RFB PORT # PROCESS ID
:1 5901 4361
You can use the KeX client to connect to any of these displays.
$ nh -r kex &
[2] 4648
$
NetHunter KeX server sessions:
X DISPLAY # RFB PORT # PROCESS ID
:2 5902 4713
You can use the KeX client to connect to any of these displays.
$
ESC
CTRL
ALT

08:26 AM
Trash
File System
Home

Why Nethunter?

- Turn you phone into a sick cyberdeck(phone?)
- Plug in usb devices for more options (SDR, more Wifi, etc)
- Phones are everywhere and who would expect a phone to be a security risk-it looks nothing like a Raspberry Pi or Flipper Zero 🐬



Need More Reasons?

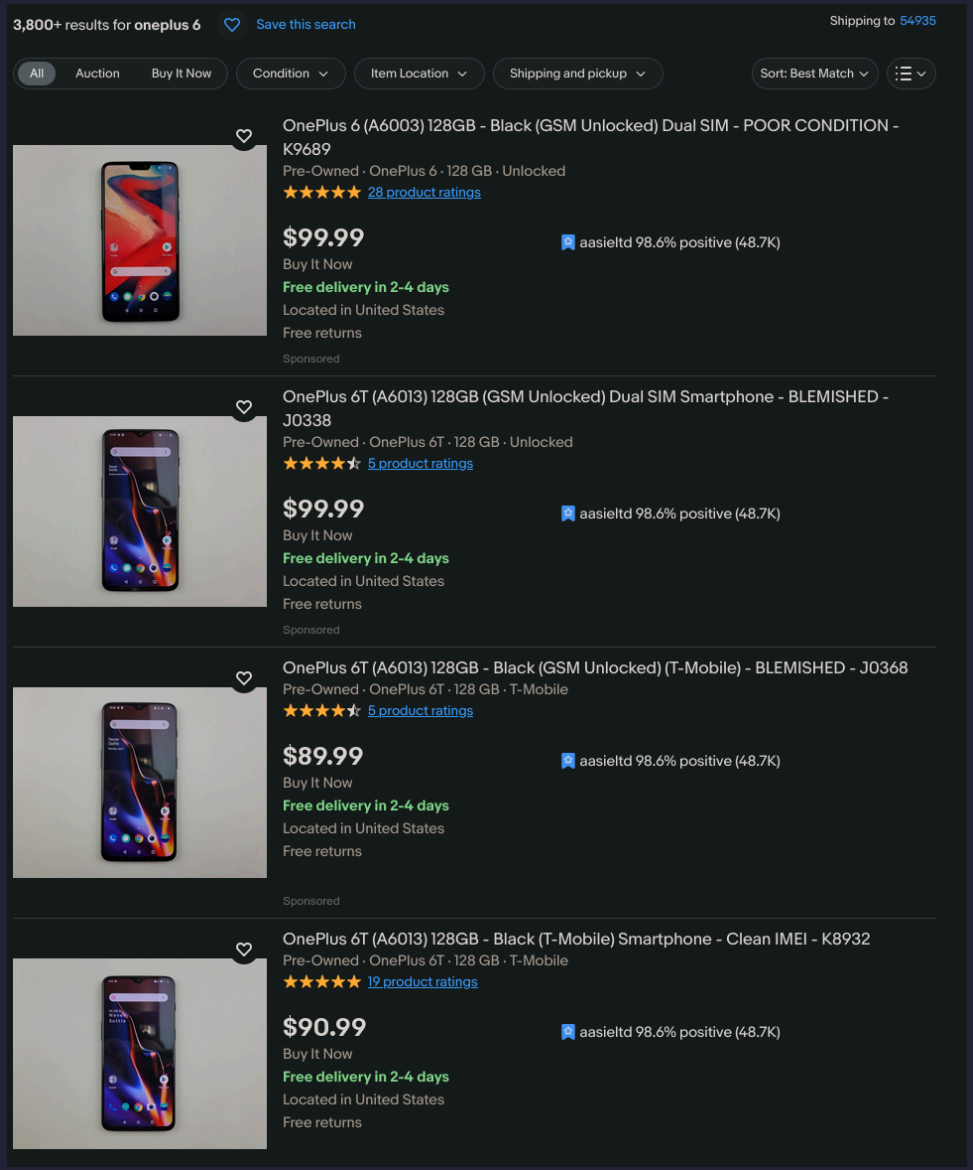
- Bad USB
- Wigle/Kismet Wardriving
- BT Attacks
- Wifi Scans, Deauth
- CAN Arsenal - Car Hacking
- It's your hardware-run what you want!

What do we need to get started?

- One or more supported devices: tablet, phone, or watch (Sorry iOS 🥲)
- A few minutes or maybe an afternoon
- Decide which edition you want to try

Where could we get a device?

- Ebay
- Facebook Marketplace
- local classifieds
- probably in that desk drawer back at home



Phone Options

Nethunter "Pro" - Run Nethunter Solo
- Pinephone and ~~PinePhone Pro~~
- Some Android Phones

Nethunter - Run Nethunter alongside Android
- Most Android Phones

Install Nethunter

https://www.kali.org/get-kali/#kali-mobile

Editions

Edition	Usage
NetHunter Lite	The full NetHunter package for rooted phones without a custom kernel.
NetHunter	The full NetHunter package with custom kernel for supported devices

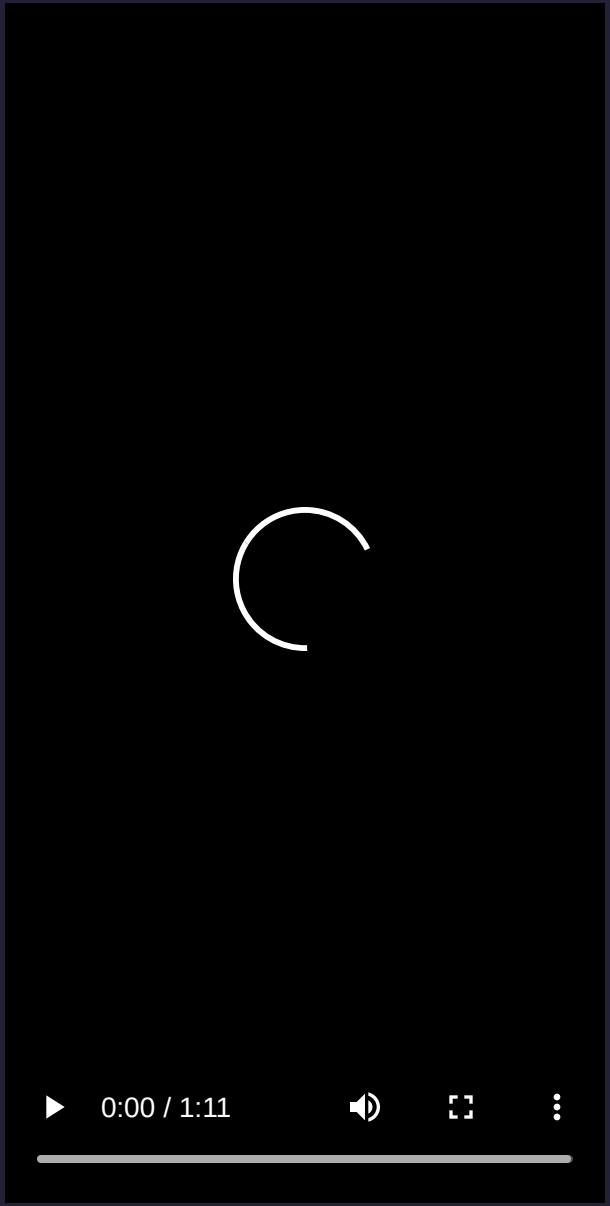
Feature	NetHunter Lite	NetHunter
App Store	Yes	Yes
Kali cli	Yes	Yes
All Kali packages	Yes	Yes
KeX	Yes	Yes
Metasploit w/o DB	Yes	Yes
Metasploit with DB	Yes	Yes
NetHunter App	Yes	Yes
Requires TWRP	Yes	Yes
Requires Root	Yes	Yes

Couple Differences

Feature	NetHunter Lite	NetHunter
WiFi Injection	No	Yes
HID attacks	No	Yes
BT Arsenal	No	Yes
CAN Arsenal	No	Yes

Probably not daily driver phone material

Pinephone Tour


0:00 / 1:11

PinePhone Impressions

Kinda like having a Raspberry Pi in your pocket-for better or worse.

The functionality is all there but the Posh Mobile interface isn't my favorite

The beta/developer hardware probably isn't doing the vision for the experience justice

Shout outs to the Pine64 IRC/Discord!

Pine64 has said they are discontinuing the Pinephone Pro - "standard" is still being sold

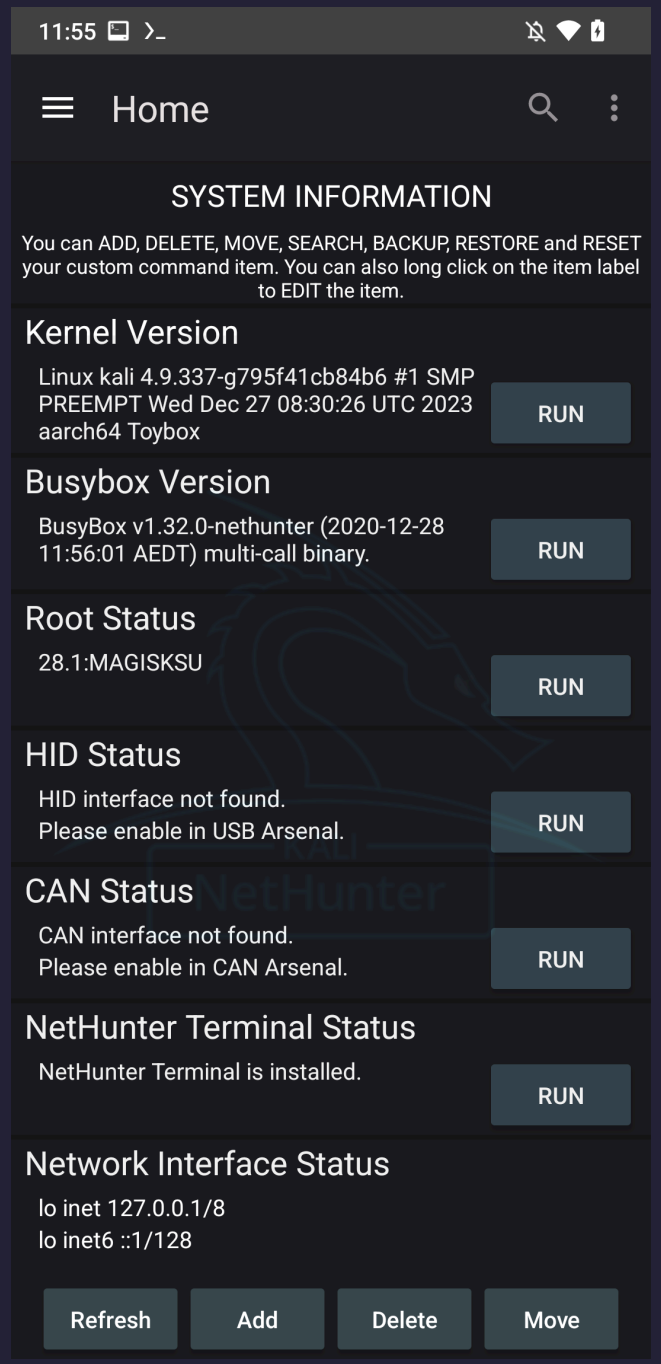
Android Impressions

Still rocking a OnePlus6 as my Nethunter device

One of the Nethunter "Pro" compatible devices

Root access lets us re-purpose phone hardware (ex. OnePlus alert slider)

Android apps are nice to have. Sideload Gapps if you need them.



Enough Slides. Do a demo already...

Is Rootless An Option?

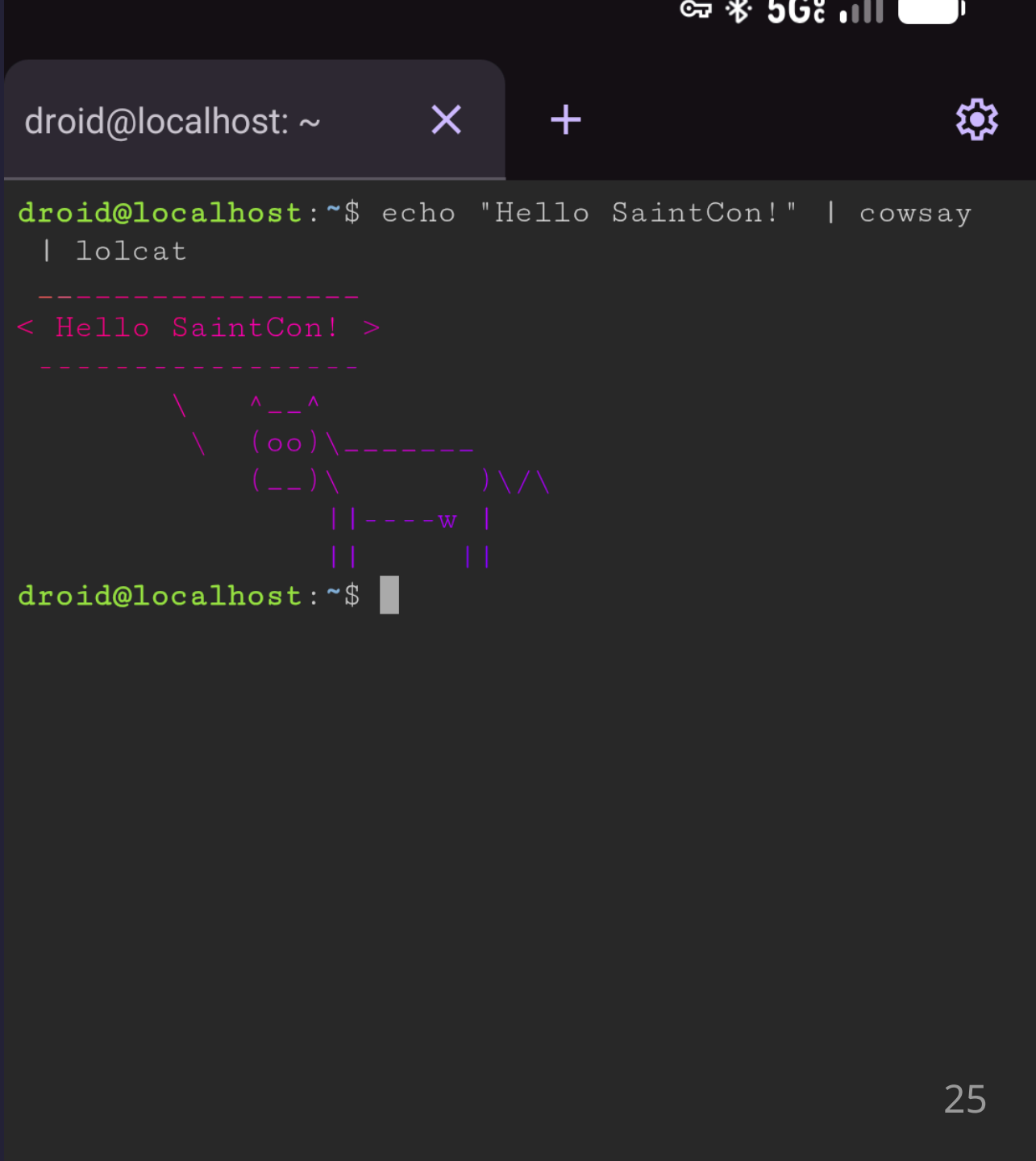
Yes, but....

NetHunter Rootless	The core of NetHunter for unrooted, unmodified devices

Feature	NetHunter Rootless
App Store	Yes
Kali cli	Yes
All Kali packages	Yes
KeX	Yes
Metasploit w/o DB	Yes

New Android Terminal

- Select Pixel or Samsung Android 16 devices - more soon?
- Full Debian VM
- Still some (Android) system and hardware restrictions
- It runs Doom (and other GUI apps)



Ideas Next Steps

- Mobile drop phone C2 node
- "Off grid" file share or webserver
- Bluetooth Payloads
- CANBUS - This is pretty new to Nethunter
- Upcycle some phones and have some fun

Q & A

Slides

Reach out and show me what you do with Nethunter!

Slides are here:
https://github.com/braydenhouston/CC-KaliNethunter/tree/saintcon-2025

Helpful Resources

https://forums.kali.org/

Pine64 - Discord and IRC channels

https://pine64.org/2025/08/14/august_2025_short_update/

@davidbombale

https://justhack.in/connect-to-kali-nethunter-vnc-server-from-local-network

Android Virtualization Framework Docs

https://source.android.com/docs/core/virtualization

Android Terminal Doom

https://www.androidauthority.com/android-16-linux-terminal-doom-3521804/